Supplemental Online Content

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1. The TRAUMOX2 Trial Group

1.1 Steering committee

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1.2 Trial statistician

Volkert Siersma.

1.3 Investigators

Listed by country and hospital.

Denmark

Hospital	Names
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Odense University Hospital	Søren Mikkelsen (primary investigator), Stine Thorhauge Zwisler, Louise Breum Petersen, Peter Martin Hansen, Søren Darling, Maria Charlotte Reiersøl Mørkeberg
Aarhus University Hospital	Mikkel Andersen (primary investigator), Christian Fenger-Eriksen, Peter Trappaud Bach

The Netherlands

Hospital	Names
Erasmus MC, University Medical Center Rotterdam	Mark G. Van Vledder (primary investigator), Esther M.M. Van Lieshout, Dennis Den Hartog, Markus Klimek, Niki A. Ottenhof, Iscander M. Maissan

Switzerland

Hospital	Names
Inselspital Bern	Wolf E. Hautz (primary investigator), Dominik A. Jacob, Roland Albrecht, Matthias Haenggi, Manuela Iten

1.4 Primary outcome assessors

Listed by country.

Denmark

Peter Martin Hansen, Johan Heiberg, Lars Simon Rasmussen, Dan Isbye.

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1.5 Data Monitoring and Safety Committee (DMSC)

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Neurobiology Research Unit and BrainDrugs, Copenhagen University Hospital, Rigshospitalet Denmark

2. Participating Prehospital Bases and Trauma Centers

2.1 Prehospital bases

2.1.1 Denmark

Physician staffed Critical Care Units (physician-staffed ambulance vehicles)

All physician staffed Critical Care Units have one referring trauma center.

Referring trauma center	Critical Care Unit
Rigshospitalet, Copenhagen University Hospital	Physician staffed Critical Care Unit 1 and 2 – City: Copenhagen (2 vehicles) Physician staffed Critical Care Unit 3 – South: Hvidovre (1 vehicle) Physician staffed Critical Care Unit 4 – North: Hilleroed (1 vehicle) Physician staffed Critical Care Unit 5 – Center: Herlev (1 vehicle)
Odense University Hospital	Physician staffed Critical Care Unit: Odense (1 vehicle)
Aarhus University Hospital	Physician staffed Critical Care Unit: Aarhus (1 vehicle) Physician staffed Critical Care Unit: Randers (1 vehicle)

Helicopter Emergency Medical Services (HEMS)

City	Air Ambulance
Ringsted	Danish Air Ambulance Ringsted
Billund	Danish Air Ambulance Billund

Skive	Danish Air Ambulance Skive
Saltum	Danish Air Ambulance Saltum

2.1.2 The Netherlands

City	Air Ambulance
Rotterdam	Lifeliner 2 Helicopter Emergency Medical Services (including a physician-staffed ambulance vehicle)

2.1.3 Switzerland

Helicopter Emergency Medical Services (HEMS)

City	Air Ambulance
Bern	REGA 3
Wilderswil	REGA 10
Zweisimmen	REGA 14

2.2 Trauma centers

Participating trauma centers should be able to provide definitive treatment of trauma patients (i.e., no transfer to more specialized institution needed), possess a trauma registry, and have a minimum average of approximately 400 trauma team activations per year.

2.2.1 Denmark

City	Trauma department
Copenhagen	Department of Anesthesia Center of Head and Orthopedics Rigshospitalet, University of Copenhagen Inge Lehmanns Vej 6, DK-2100 Copenhagen
Odense	Department of Anesthesiology and Intensive Care Odense University Hospital J. B. Winsløws Vej 4, DK-5000 Odense
Aarhus	Department of Anesthesiology Aarhus University Hospital Palle Juul-Jensens Boulevard 99, DK-8200 Aarhus

2.2.2 The Netherlands

City	Trauma department
Rotterdam	Emergency Department Erasmus MC, University Medical Center Rotterdam P.O. Box 2040, 3000 CA Rotterdam

2.2.3 Switzerland

City	Trauma department
Bern	Department of Emergency Medicine Inselspital University Hospital Bern Rosenbühlgasse 27, 3010 Bern

3. Supplementary Methods

3.1 Research ethics committee/IRB approvals

Country	Ethics approval
Denmark	The Regional Scientific Ethical Committee, approval no. H-21018062 First approved October 20, 2021 The approval was given to all participating sites in Copenhagen, Odense, and Aarhus with the coordinating site in Copenhagen as the applicant
The Netherlands	Medical Research Ethics Committee Erasmus MC, approval no. MEC-2021 First approved February 24, 2022
Switzerland	Ethics Committee of the Canton Bern, approval no. 2022-01335 First approved November 1, 2022

3.2 Details on intervention data

The arterial oxyhemoglobin saturation measured by pulse oximetry (SpO_2),the type of oxygen delivery, flow (liters of oxygen per minute), and fraction of inspired oxygen (FiO_2) values were collected and registered hourly. In case of missing values on the data collection sheet, an approximated hourly median value of SpO_2, flow and FiO_2 were used from the medical record. The use of high-flow nasal cannula was noted on the data collection sheet as a non-rebreather mask. In case more arterial blood gasses (ABGs) were obtained in the intervention period besides the protocolized two ABGs at 1 hour ± 30 minutes and 6 hours ± 2 hours after randomization, the results from the ABG closest to the designated time points were used.

3.3 Details on outcomes

3.3.1 Primary outcome and key secondary outcomes

The primary outcome was a composite of death and/or major respiratory complications within 30 days. Major respiratory complications were defined as pneumonia based on the Centers for Disease Control and Prevention (CDC) criteria[1] and/or acute respiratory distress syndrome (ARDS) based on the Berlin definition.[2] Assessment of the primary outcome was blinded for the assessors. All primary outcome assessors were asked to guess the treatment allocation in their assessments. The key secondary outcomes were death and major respiratory complications within 30 days individually.

Death

Investigators registered death within 30 days using the medical record or national/local registries. If that was not possible, for instance if the patient was a foreigner, death within 30 days was assessed by contacting the patient or relatives by phone or e-mail at day 30 or after from the day of inclusion. At least three attempts on three different days were carried out. If no contact was obtained, and the patient was discharged to home from either the TRAUMOX2 participating site or subsequent receiving hospital, the patient was considered to be alive on day 30. However, if the patient was transferred to another hospital without medical record access for investigators, 30-day vital status was considered as missing data.

Major respiratory complications

The blinded assessment of pneumonia and/or ARDS within 30 days from inclusion was based on the available medical record system. If the patient was transferred to another hospital, and the local medical record until day 30 was not available, the assessment was based on the available medical record from the TRAUMOX2 participating site only. However, an attempt of obtaining the medical record of the receiving facility was carried out and included in the primary outcome assessment if possible. Pneumonia and ARDS diagnoses were accompanied by dates of event. Pneumonia was categorized as non-ventilator associated pneumonia (PNEU) or ventilator-associated pneumonia (VAP); please see the criteria below. ARDS was categorized as mild, moderate, or severe; please see the criteria below.

A shortened version of the pneumonia CDC criteria is shown below. Please see the reference for full details.

<table>
<tr><th colspan="2" align="left">Pneumonia CDC criteria[1]</th></tr>
<tr><td colspan="2">Pneumonia (PNEU) is identified by using a combination of imaging, clinical and laboratory criteria.

Date of event: For a PNEU/VAP the date of event is the date when the first element used to meet the PNEU infection criterion occurred for the first time within the 7-day Infection Window Period.

Ventilator-associated pneumonia (VAP): A pneumonia where the patient is on mechanical ventilation for >2 calendar days on the date of event, with day of ventilator placement being Day 1,*
AND
the ventilator was in place on the date of event or the day before.
*If the ventilator was in place prior to inpatient admission, the ventilator day count begins with the admission date to the first inpatient location.</td></tr>
<tr><th>Imaging Test Evidence</th><th>Signs/Symptoms</th></tr>
<tr><td>Two or more serial chest imaging test results with at least one of the following:

New and persistent
or
Progressive and persistent
• Infiltrate
• Consolidation
• Cavitation

Note: In patients without underlying pulmonary or cardiac disease (for example: respiratory distress syndrome, bronchopulmonary dysplasia, pulmonary edema, or chronic obstructive pulmonary disease), one definitive imaging test result is acceptable.</td><td>For ANY PATIENT, at least <u>one</u> of the following:

• Fever (>38.0°C or >100.4°F)
• Leukopenia (≤4000 WBC/mm3) or leukocytosis (≥12,000 WBC/mm3)
• For adults ≥70 years old, altered mental status with no other recognized cause

And at least <u>two</u> of the following:

• New onset of purulent sputum or change in character of sputum, or increased respiratory secretions, or increased suctioning requirements
• New onset or worsening cough, or dyspnea, or tachypnea
• Rales or bronchial breath sounds
• Worsening gas exchange (for example: O_2 desaturations (for example: $PaO_2/FiO_2 \leq 240$ mmHg), increased oxygen requirements, or increased ventilator demand)</td></tr>
</table>

Table 3 from the ARDS reference is shown below. Please see the reference for full details.

The Berlin Definition of Acute Respiratory Distress Syndrome[2]	
Timing	Within 1 week of a known clinical insult or new or worsening respiratory symptoms
Chest imaging[a]	Bilateral opacities—not fully explained by effusions, lobar/lung collapse, or nodules
Origin of edema	Respiratory failure not fully explained by cardiac failure or fluid overload Need objective assessment (eg, echocardiography) to exclude hydrostatic edema if no risk factor present
Oxygenation[b]	Mild: 200 mmHg < $PaO_2/FIO_2 \leq$ 300 mmHg with PEEP or CPAP $\geq$5cmH_2O[c]
	Moderate: 100 mmHg < $PaO_2/FIO_2 \leq$ 200 mmHg with PEEP $\geq$5cmH_2O
	Severe: $PaO_2/FIO_2 \leq$ 100 mmHg with PEEP $\geq$5cmH_2O
Abbreviations: CPAP, continuous positive airway pressure; FIO_2, fraction of inspired oxygen; PaO_2, partial pressure of arterial oxygen; PEEP, positive end-expiratory pressure. [a]Chest radiograph or computed tomography scan. [b]If altitude is higher than 1000 m, the correction factor should be calculated as follows: [PaO_2/FIO_2 x (barometric pressure/760)]. [c]This may be delivered noninvasively in the mild acute respiratory distress syndrome group.	

3.3.2 Exploratory outcomes

Only exploratory outcomes that required further elaboration than what was stated in the statistical analysis plan is mentioned in the table below.

Outcome	**Definition**
Episode(s) of hypoxemia during the eight-hour intervention	Number of times the hourly collected SpO_2 value from the data collection sheet was <90%
Surgical site infection(s) within 30 days	At least one of the following stated in the medical record (based on the CDC criteria for a surgical site event[3]): • Purulent drainage • Positive microbiologic testing from surgical site/wound • Deliberately opened by a surgeon AND sign(s) of infection • Diagnosis of surgical site infection by a surgeon
Ventilator hours within 30 days	Ventilator hours was defined as: Positive pressure ventilation at any time, including Bilevel Positive Airway Pressure (BiPAP) The following treatments were not considered as mechanical ventilation: Non-invasive ventilation (NIV), high flow nasal cannula or Continuous Positive Airway Pressure (CPAP) Endotracheal intubation and mechanical ventilation during surgery were not included in the above definitions if the patient was breathing spontaneously without an endotracheal tube before surgery.

	Only mechanical ventilation started pre-hospital, in the trauma center or in the ICU was considered. Hour(s) were calculated as commenced hour(s) with no decimals (eg, 1 hour and 15 minutes on mechanical ventilation = 2 hours on mechanical ventilation)
Ventilator-free days within 30 days	The definition for being ventilated in the above was also applied here for calculating ventilator-free days. Days were calculated as commenced day(s) with no decimals (eg, 1 hour on mechanical ventilation = 1 day = 29 days ventilator-free days)
Days alive outside the ICU within 30 days	Day(s) were calculated as commenced day(s) with no decimals (eg, 1 hour of ICU admission = 1 ICU day = 29 days alive outside the ICU)
ICU length of stay (ICU LOS)	Calculated as commenced day(s) with no decimals (eg, 25 hours of ICU admission = 2 ICU days)
Hospital length of stay (HOS LOS)	Calculated as commenced day(s) with no decimals (eg, 25 hours of admission = 2 admission days) Only the primary admission was considered (no re-admissions). If a patient was transferred from one hospital to another hospital for further treatment related to the trauma, it was not considered a hospital discharge. If a patient was transferred from the hospital to a psychiatry department or a rehabilitation center for further treatment, then the day of transfer was considered the date of hospital discharge
Sepsis during hospital admission	Clearly stated in the patient's medical record by a physician or assigned as a diagnosis
Pneumonia post-discharge within 30 days	Depending on the site: Evaluated through drugs prescribed after hospital discharge in countries where this information was available or by calling the patients at the 6-month follow-up

We are still collecting long-term follow-up data at 6 and 12 months on functional outcome (GOSE) and quality of life (EQ-5D-5L). This will be presented later in the follow-up study. A statistical analysis plan for handling these data will become available on our trial website www.traumox2.org beforehand.

3.4 Adverse Events (AEs), Serious Adverse Events (SAEs) and Suspected Unexpected Serious Adverse Reactions (SUSARs)

We defined Adverse Events (AEs) as any untoward medical occurrence in a subject to whom a medicinal product is administered, and does not necessarily have a causal relationship with this treatment. This group of trial patients were expected to have a lot of complications due to trauma. It is the established practice in trials on critically ill patients that adverse events are part of the natural trajectory of the primary disease process or expected complications of the critical illness.[4] Therefore, we chose to record only the following until discharge (but a maximum of 30 days) as AEs:

- Atelectasis (only atelectases assessed by a radiologist were recorded)
- Irritability of airway mucosa (recorded only if registered by the health care staff in the medical record)

Only the first AE of either atelectasis or irritability of airway mucosa was recorded during admission (not a subsequent).

We defined Serious Adverse Events (SAEs) as any untoward medical occurrence that resulted in death, was life-threatening, involved or prolonged hospital length of stay (eg, re-admission), involved persistence of significant disability or incapacity or resulted in a congenital anomaly or birth defect.

To monitor AEs and SAEs, a TRAUMOX2 investigator would assess the trial participant's medical record:

- Once within the first 24 hours
- Every third day until discharge (maximum of 30 days)

Suspected Unexpected Serious Adverse Reactions (SUSARs) were not expected in this trial given the long-standing history for the use of supplemental oxygen in medicine for over a century with its safety profile being well-investigated with no major safety concerns.

3.5 Major protocol violations of oxygenation

All patients with ≥1 major protocol violation of oxygenation were removed in the per-protocol analyses. The per-protocol population was the same as the modified intention-to-treat population with exclusion of the patients having one or more major protocol violations.

A major protocol violation of oxygenation during the eight-hour intervention was defined as:

Restrictive oxygen group	
Non-intubated patients	**Intubated patients**
Supplemental oxygen ≥3 liters of oxygen per minute AND SpO_2 ≥98% recorded at two consecutive hourly time points on the data collection sheet	FiO_2 >0.4 AND SpO_2 ≥98% recorded at two consecutive hourly time points on the data collection sheet

Liberal oxygen group	
Non-intubated patients	**Intubated patients**
Supplemental oxygen <3 liters of oxygen per minute at two consecutive hourly time points on the data collection sheet	FiO_2 <0.4 at two consecutive hourly time points on the data collection sheet

3.6 Protocol deviations

For all TRAUMOX2 patients, it was possible to deviate from the protocol if clinically justified by the treating physician. Such deviations were always documented including the clinical justification in our database.

3.7 Details on statistical analyses

3.7.1 Blinding while analyzing

The trial statistician was blinded to the treatment allocation, and the manuscript authors became blinded to the treatment allocation once the trial concluded and the data were exported for analysis. In the exported dataset, the randomization groups were renamed to eg, "GR12" and "GR17", by an unblinded research assistant who was not otherwise involved in the trial. Subsequently, the renamed dataset was provided to the trial statistician for data analysis.

3.7.2 Handling of missing outcome data

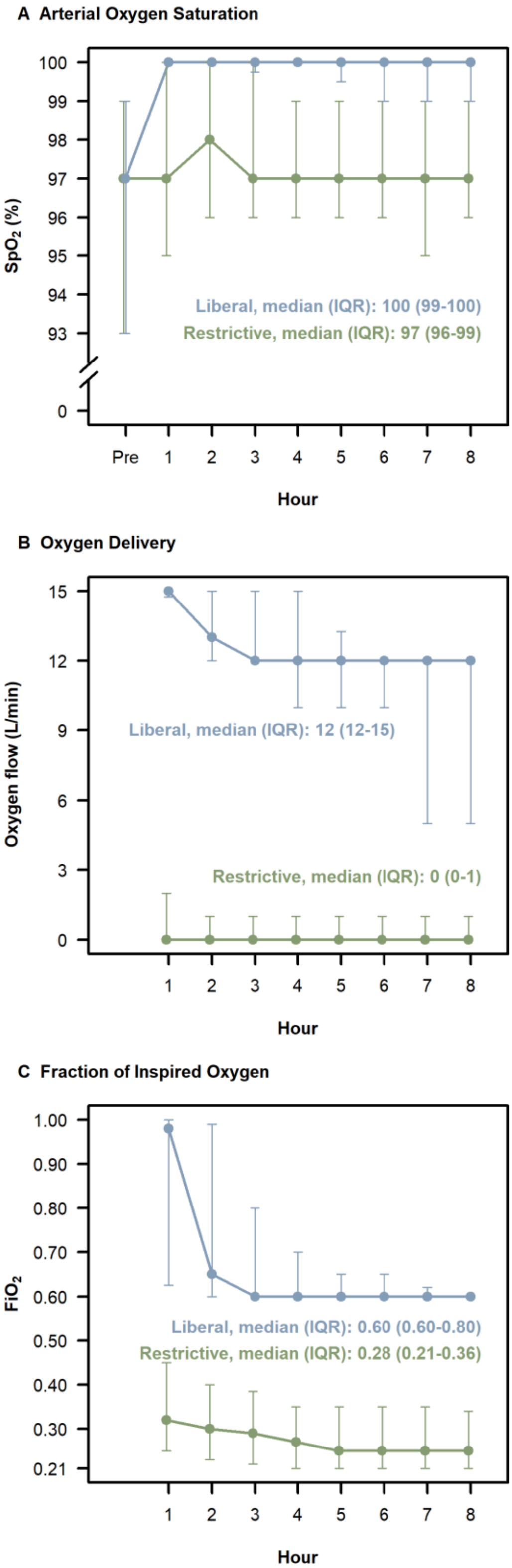
If patients withdrew their consent once they regained consciousness, data up until that day were collected and stored. However, the participating sites' individual approach were always according to national law. If data were available for time-dependent outcomes (eg, outcomes that were specified to be assessed 7 or 30 days after inclusion), the assessments were done dependent on the timing of withdrawal of consent. In case a time-dependent outcome occurred within the period of available data for patients who withdrew consent, the outcome event was registered. If an outcome did not occur within the period of available data for a time-dependent outcome, the outcome was registered as missing data.

Missing outcome data, which covered patients who withdrew consent or were unavailable for follow-up, was adjusted for using inverse probability weighting.[5] In this method, patients with available outcome data were weighted by the inverse of the probability of being observed. Specifically for the primary outcome, if major respiratory complications were not present before the consent was withdrawn, we considered both death and major respiratory complications within 30 days to be missing. The weights were calculated in a logistic regression model based on the allocation, the dominating injury type, the type of oxygen supply, psychiatric comorbidity and (for Danish patients) whether a personal identification number was registered, or the use of a temporary personal identification number. Adjustment of the variance matrix to reflect the weighting was included in the generalized estimating equations method.

4. Supplementary Results

4.1 eFigure

eFigure. Oxygen Intervention Characteristics



Trauma patients allocated to either a restrictive or liberal oxygen strategy. The “Pre” term in panel A refers to the first SpO_2 measurement from the prehospital setting. SpO_2: Arterial oxygen saturation measured by pulse oximetry. L/min: Liters per minute. FiO_2: Fraction of inspired oxygen.

4.2 eTables

eTable 1. Additional Characteristics of Patient Enrollment Location and Receiving Trauma Center		
Characteristics	**Restrictive oxygen group (N=750)**	**Liberal oxygen group (N=758)**
Enrollment location – no./total no. (%)		
Prehospital		
Physician staffed Critical Care Unit 1 and 2 – City: Copenhagen, Denmark	62/750 (8.3)	68/758 (9.0)
Physician staffed Critical Care Unit 3 – South: Hvidovre, Denmark	25/750 (3.3)	21/758 (2.8)
Physician staffed Critical Care Unit 4 – North: Hilleroed, Denmark	25/750 (3.3)	26/758 (3.4)
Physician staffed Critical Care Unit 5 – Center: Herlev, Denmark	34/750 (4.5)	36/758 (4.7)
Danish Air Ambulance Ringsted, Denmark	40/750 (5.3)	37/758 (4.9)
Danish Air Ambulance Billund, Denmark	21/750 (2.8)	23/758 (3.0)
Danish Air Ambulance Skive, Denmark	17/750 (2.3)	18/758 (2.4)
Danish Air Ambulance Saltum, Denmark	2/750 (0.3)	2/758 (0.3)
Physician staffed Critical Care Unit: Odense, Denmark	38/750 (5.1)	39/758 (5.1)
Physician staffed Critical Care Unit: Aarhus, Denmark	13/750 (1.7)	13/758 (1.7)
Physician staffed Critical Care Unit: Randers, Denmark	1/750 (0.1)	0/758 (0.0)
Lifeliner 2 Helicopter Emergency Medical Services (including a physician-staffed ambulance vehicle), Rotterdam, The Netherlands	5/750 (0.7)	3/758 (0.4)
REGA 3, Swiss-Air Ambulance, Bern, Switzerland	19/750 (2.5)	16/758 (2.1)
REGA 10, Swiss-Air Ambulance, Wilderswil, Switzerland	3/750 (0.4)	3/758 (0.4)
REGA 14, Swiss-Air Ambulance, Zweisimmen, Switzerland	3/750 (0.4)	2/758 (0.3)
In-hospital		
Rigshospitalet, Copenhagen University Hospital, Denmark	150/750 (20.0)	160/758 (21.1)
Odense University Hospital, Odense, Denmark	118/750 (15.7)	117/758 (15.4)
Aarhus University Hospital, Aarhus, Denmark	62/750 (8.3)	58/758 (7.7)
Erasmus MC, University Medical Center Rotterdam, Rotterdam, The Netherlands	87/750 (11.6)	90/758 (11.9)
University Hospital Bern, Switzerland	25/750 (3.3)	26/758 (3.4)
Receiving trauma center – no./total no. (%) (N=1508)		
Rigshospitalet, Copenhagen University Hospital, Denmark	335/750 (44.4)	348/758 (45.7)
Odense University Hospital, Odense, Denmark	173/750 (22.9)	177/758 (23.2)
Aarhus University Hospital, Aarhus, Denmark	100/750 (13.3)	93/758 (12.2)
Erasmus MC, University Medical Center Rotterdam, Rotterdam, The Netherlands	92/750 (12.2)	93/758 (12.2)
University Hospital Bern, Switzerland	50/750 (7.2)	47/758 (6.7)

Trauma patients allocated to either a restrictive or liberal oxygen strategy.

eTable 2. Additional Characteristics of Patients		
Characteristics	**Restrictive oxygen group (N=750)**	**Liberal oxygen group (N=758)**
Body Mass Index – kg/m^2	25.0 (22.5-28.1) [n=628]	25.0 (22.8-27.8) [n=615]
Mechanism of injury – no./total no. (%)		
Traffic: Motor vehicle accident	147/742 (19.8)	151/756 (20.0)
Traffic: Motorcycle accident	74/742 (10.0)	70/756 (9.3)
Traffic: Bicycle accident	109/742 (14.7)	108/756 (14.3)
Traffic: Pedestrian	53/742 (7.1)	50/756 (6.6)
Traffic: Other (eg, ship, airplane, or railway train)	14/742 (1.9)	19/756 (2.5)
Shot by handgun, shotgun, rifle or other firearm of any caliber	8/724 (1.1)	6/756 (0.8)
Stabbed by knife, sword, dagger, other pointed or sharp object	69/742 (9.3)	68/756 (9.0)
Struck or hit by blunt object	51/742 (6.9)	58/756 (7.7)
Fall: 0-2 meters	82/742 (11.1)	69/756 (9.1)
Fall: 2-4 meters	66/742 (8.9)	68/756 (9.0)
Fall: >4 meters	58/742 (7.8)	57/756 (7.5)
Blast/explosion	3/742 (0.4)	4/756 (0.5)
Other	8/742 (1.1)	28/756 (3.7)
Type of transport to the trauma center – no./total no. (%)		
Ground ambulance	583/747 (78.0)	585/743 (78.7)
Helicopter ambulance	140/747 (18.7)	141/743 (19.0)
A combination of ground ambulance and helicopter ambulance	14/747 (1.9)	9/743 (1.2)
Private vehicle	1/747 (0.1)	3/743 (0.4)
Walk-in	9/747 (1.2)	4/743 (5.3)
Other	0/747 (0.0)	1/743 (0.1)
Use of prehospital or in-hospital supplemental oxygen prior to randomization – no./total no. (%)	333/713 (46.7)	351/718 (48.9)
Highest SpO_2 measured prior to randomization – %	100 (97-100) [n=265]	99 (97-100) [n=285]
Body temperature measured on admission – ^{o}C	36.5 (35.9-37.0) [n=479]	36.5 (35.9-36.9) [n=484]
Surgery performed in the trauma resuscitation room		
Neurosurgery – no./total no. (%)	3/744 (0.4)	1/747 (0.1)
Cardiothoracic surgery – no./total no. (%)	10/744 (1.3)	9/747 (1.2)
Abdominal surgery – no./total no. (%)	0/744 (0.0)	4/747 (0.5)
Orthopedic surgery – no./total no. (%)	2/744 (0.3)	1/747 (0.1)
Vascular surgery – no./total no. (%)	1/744 (0.1)	0/747 (1.1)
Other(s) – no./total no. (%)	1/744 (0.1)	1/747 (0.1)

Trauma patients allocated to either a restrictive or liberal oxygen strategy. Continuous variables are presented as median with interquartile range.

eTable 3. Oxygen Intervention Characteristics		
Characteristic	**Restrictive oxygen group (N=750)**	**Liberal oxygen group (N=758)**
Time from trauma to randomization – min (N=1268)	54 (35-79) [n = 633]	52 (33-78) [n=635]
Type of supplemental oxygen* – no./total no. (%)		
No supplemental oxygen	200/739 (27.1)	13/735 (1.8)
Nasal cannula	171/739 (23.1)	45/735 (6.1)
Non-rebreather mask	18/739 (2.4)	340/735 (46.2)
Intubated (at any point during the intervention period)	350/739 (47.4)	337/735 (45.9)
Hemoglobin hour 1 ± 30 min § – g/dL	12.9 (11.4-14.0) [n=602]	12.7 (11.3-14.0) [n=603]
Hemoglobin hour 6 ± 2 hours § – g/dL	12.5 (10.8-13.7) [n=482]	12.4 (10.6-13.9) [n=487]
Lactate hour 1 ± 30 min § – mmol/L	1.5 (0.9-2.5) [n=585]	1.4 (0.9-2.4) [n=590]
Lactate hour 6 ± 2 hours § – mmol/L	1.4 (0.9-2.2) [n=472]	1.4 (0.9-2.1) [n=479]
Major protocol violation ¶ – no./total no. (%)	50/742 (6.7)	102/744 (13.7)
Protocol deviation † – no./total no. (%)	40/741 (5.4)	17/738 (2.3)

Trauma patients allocated to either a restrictive or liberal oxygen strategy. The time from trauma to randomization is presented with a median and interquartile range.
* The data collection sheet consisted of eight hourly data points (H1-H8). The sum of no supplemental oxygen, nasal cannula, non-rebreather mask or intubated is 100%. Patients are in the no supplemental oxygen category if none of the eight data points were nasal cannula, non-rebreather mask or intubated. If just one of the eight data points was intubated, the patient was categorized as intubated. Patients could receive both supplemental oxygen via a nasal cannula and non-rebreather mask during the 8-hour intervention period at different time points and were registered in the appropriate category based on the majority of the data points from the sheet. In case they were evenly distributed, the patient was registered in the non-rebreather mask category.
§ The 1st and 2nd arterial blood gases during the oxygen intervention were obtained at hour 1 ± 30 min after randomization and hour 6 ± 2 hours after randomization, respectively.
¶ See page 10 for definitions of major protocol violations.
† See page 10 for definition of protocol deviations.

eTable 4. Primary and Key Secondary Outcomes

	Restrictive oxygen group	Liberal oxygen group	Odds ratio (95% CI)	*P* value	Adjusted odds ratio (95% CI)	*P* value	Adjusted odds ratio + smoking (95% CI)	*P* value
Primary outcome								
Death and/or major respiratory complications – no./total no. (%)	118/733 (16.1)	121/724 (16.7)	1.01 (0.75 to 1.37)	0.94	0.98 (0.68 to 1.41)	0.92	1.05 (0.73 to 1.49)	0.80
Key secondary outcomes								
Death – no./total no. (%)	63/733 (8.6)	53/724 (7.3)	1.28 (0.85 to 1.92)	0.23	1.37 (0.80 to 2.38)	0.25	1.37 (0.80 to 2.33)	0.26
Major respiratory complications – no./total no. (%)	65/733 (8.9)	78/724 (10.8)	0.84 (0.59 to 1.19)	0.33	0.84 (0.57 to 1.23)	0.38	0.90 (0.61 to 1.32)	0.58

Trial outcomes for trauma patients allocated to either a restrictive or liberal oxygen strategy. Death and major respiratory complications were both evaluated within 30 days. The smoking variable reflects only active smokers, identified from medical records and, in some cases, confirmed directly with patients.

eTable 5. Categorical Exploratory Outcomes

	Restrictive oxygen group	Liberal oxygen group	Odds ratio (95% CI)	*P* value	Adjusted odds ratio (95% CI)	*P* value	Adjusted odds ratio + smoking (95% CI)	*P* value
Presence of any hypoxemic episodes (SpO_2 <90%) * – no./total no. (%)	44/737 (6.0)	28/737 (3.8)	1.67 (1.02 to 2.70)	0.04	1.72 (1.04 to 2.78)	0.03	1.69 (1.04 to 2.78)	0.04
ICU re-admission(s) – no./total no. (%) †	17/368 (4.6)	18/352 (5.1)	0.59 (0.26 to 1.37)	0.22	0.60 (0.25 to 1.43)	0.25	0.57 (0.25 to 1.32)	0.19
Sepsis during hospital admission – no./total no. (%)	19/736 (2.6)	31/727 (4.3)	0.55 (0.30 to 1.02)	0.06	0.53 (0.29 to 1.00)	0.05	0.52 (0.27 to 1.00)	0.05
Surgical site infection(s) within 30 days – no./total no. (%)	23/736 (3.1)	32/724 (4.4)	0.50 (0.25 to 0.99)	0.05	0.49 (0.26 to 0.93)	0.03	0.51 (0.27 to 0.96)	0.04
Pneumonia post-discharge within 30 days after enrollment ¶ – no./total no. (%)	27/640 (4.2)	24/620 (3.9)	1.06 (0.60 to 1.85)	0.86	1.04 (0.59 to 1.85)	0.89	1.03 (0.57 to 1.85)	0.93
Brain injury within 7 days of admission – no./total no. (%)	290/738 (39.3)	275/732 (37.6)	1.11 (0.88 to 1.41)	0.35	1.05 (0.82 to 1.35)	0.68	1.04 (0.81 to 1.33)	0.74
Myocardial infarction within 7 days – no./total no. (%)	0/738 (0.0)	1/730 (0.1)	-	-	-	-	-	-
Cerebral ischemia within 7 days – no./total no. (%)	16/738 (2.2)	10/730 (1.4)	1.56 (0.68 to 3.57)	0.30	1.45 (0.63 to 3.45)	0.38	1.47 (0.64 to 3.45)	0.36
ICU admission – no./total no. (%)	374/738 (50.7)	359/736 (48.8)	1.16 (0.90 to 1.49)	0.24	1.01 (0.76 to 1.33)	0.95	1.01 (0.76 to 1.33)	0.95

Trial outcomes for trauma patients allocated to either a restrictive or liberal oxygen strategy. No exploratory outcomes were statistically significant between the groups after adjusting for multiple testing. The smoking variable reflects only active smokers, identified from medical records and, in some cases, confirmed directly with patients.
* Defined as any hypoxemic episodes (SpO_2 <90%) during the eight-hour intervention from the hourly collected SpO_2 values.
† ICU: Intensive care unit.
¶ Only patients discharged within 30 days were eligible for evaluation of pneumonia post-discharge.

eTable 6. Continuous Exploratory Outcomes

	Restrictive oxygen group	Liberal oxygen group	Mean difference (95% CI)	*P* value	Adjusted mean difference (95% CI)	*P* value	Adjusted mean difference + smoking (95% CI)	*P* value
Initial surgery duration after trauma resuscitation room treatment – min	114 (64-166) [n=165]	113 (68-149) [n=161]	10.3 (-8.8 to 29.3)	0.29	12.6 (-7.4 to 32.6)	0.22	15.4 (-4.3 to 35.0)	0.13
Time on mechanical ventilation – hours	0 (0-14) [n=735]	0 (0-12) [n=725]	4.8 (-7.6 to 17.2)	0.45	3.3 (-8.7 to 15.2)	0.59	3.7 (-8.1 to 15.6)	0.54
Days alive without mechanical ventilation – days	30 (29-30) [n=735]	30 (29-30) [n=723]	-0.6 (-1.4 to 0.2)	0.16	-0.4 (-1.1 to 0.3)	0.27	-0.5 (-1.2 to 0.2)	0.20
Intensive Care Unit Length of Stay – days	2 (1-9) [n=368]	2 (1-7) [n=357]	1.1 (-0.2 to 2.5)	0.10	1.0 (-0.4 to 2.3)	0.17	1.0 (-0.4 to 2.5)	0.16
Days alive outside the Intensive Care Unit within 30 days – days	29 (26-30) [n=736]	30 (27-30) [n=724]	-0.8 (-1.7 to 0.1)	0.09	-0.5 (-1.3 to 0.2)	0.17	-0.6 (-1.3 to 0.2)	0.12
Hospital Length of Stay – days	7 (3-15) [n=736]	7 (3-15) [n=728]	-0.7 (-2.3 to 0.9)	0.41	-1.0 (-2.3 to 0.6)	0.23	-1.0 (-2.5 to 0.6)	0.23

Trial outcomes for trauma patients allocated to either a restrictive or liberal oxygen strategy. All continuous variables are presented as medians with interquartile ranges in the group allocation columns. No exploratory outcomes were statistically significant between the groups after adjusting for multiple testing. The smoking variable reflects only active smokers, identified from medical records and, in some cases, confirmed directly with patients.
* Patients still admitted at day 30 were not eligible for evaluation of pneumonia post-discharge.

eTable 7. Adjusted Subgroup Analyses				
	Restrictive oxygen group	**Liberal oxygen group**	**Odds ratio (95% CI)**	**Adjusted odds ratio (95% CI)**
	no. of events/total no. of patients (%)			
All patients	118/733 (16.1)	121/724 (16.7)	1.01 (0.75 to 1.37)	0.98 (0.68 to 1.39)
Intubated at randomization				
No	56/558 (10.0)	50/543 (9.2)	1.12 (0.75 to 1.67)	1.16 (0.73 to 1.85)
Yes	62/175 (35.4)	71/181 (39.2)	0.90 (0.57 to 1.41)	0.76 (0.45 to 1.30)
ICU admission				
No	42/458 (9.2)	50/481 (10.4)	0.98 (0.64 to 1.49)	0.89 (0.56 to 1.43)
Yes	74/271 (27.3)	68/238 (28.6)	0.96 (0.63 to 1.49)	0.98 (0.58 to 1.67)
Moderate or severe traumatic brain injury *				
AIS < 3	45/500 (9.0)	48/473 (9.2)	1.06 (0.68 to 1.64)	1.11 (0.68 to 1.85)
AIS ≥ 3	73/233 (31.3)	73/202 (36.1)	0.81 (0.53 to 1.25)	0.83 (0.51 to 1.37)
Known lung disease				
No	97/667 (14.5)	105/643 (16.3)	0.88 (0.65 to 1.22)	0.95 (0.65 to 1.39)
Yes	19/59 (32.2)	14/69 (20.2)	1.85 (0.73 to 4.55)	1.06 (0.40 to 2.86)
Site of inclusion				
Prehospital	48/303 (15.8)	54/296 (18.2)	0.81 (0.51 to 1.28)	0.95 (0.55 to 1.64)
In-hospital	70/430 (16.3)	67/428 (15.7)	1.19 (0.81 to 1.79)	0.98 (0.63 to 1.52)
Injury severity score †				

	Restrictive oxygen group	Liberal oxygen group	Odds ratio (95% CI)	Adjusted odds ratio (95% CI)
	no. of events/total no. of patients (%)			
ISS ≤ 15	22/380 (5.8)	19/389 (4.9)	1.33 (0.69 to 2.56)	1.14 (0.56 to 2.33)
ISS > 15	95/352 (27.0)	100/329 (30.4)	0.86 (0.60 to 1.23)	0.93 (0.62 to 1.40)

Adjusted subgroup analyses for trauma patients allocated to either a restrictive or liberal oxygen strategy.
* Moderate or severe traumatic brain injury was evaluated by a score of 3 or higher on the AIS (Abbreviated Injury Scale) from the head/neck codes, but only head codes were selected.
† Injury Severity Score: ISS.

eTable 8. Per-Protocol Analyses of Primary and Key Secondary Outcomes ¶								
	Restrictive oxygen group (N=700)	**Liberal oxygen group (N=656)**	**Odds ratio (95% CI)**	***P* value**	**Adjusted odds ratio (95% CI)**	***P* value**	**Adjusted odds ratio + smoking (95% CI)**	***P* value**
Primary outcome								
Death and/or major respiratory complications – no./total no. (%)	104/683 (15.2)	105/627 (16.7)	0.98 (0.71 to 1.35)	0.91	0.97 (0.67 to 1.41)	0.87	1.04 (0.71 to 1.52)	0.83
Key secondary outcomes								
Death – no./total no. (%)	54/683 (7.9)	44/627 (7.0)	1.27 (0.82 to 2.00)	0.29	1.25 (0.70 to 2.22)	0.45	1.22 (0.69 to 2.17)	0.49
Major respiratory complications – no./total no. (%)	58/685 (8.5)	69/629 (11.0)	0.78 (0.54 to 1.14)	0.20	0.82 (0.54 to 1.23)	0.34	0.89 (0.59 to 1.33)	0.58

Per-protocol analyses of trauma patients allocated to either a restrictive or liberal oxygen strategy. Death and major respiratory complications were both evaluated within 30 days. The smoking variable reflects only active smokers, identified from medical records and, in some cases, confirmed directly with patients.
¶ See page 10 for definitions of major protocol violations.

eTable 9. Adverse Events (AE) and Serious Adverse Events (SAE) ¶		
	Restrictive oxygen group (N=750)	**Liberal oxygen group (N=758)**
Adverse event (AE) – no./total no. (%)		
Atelectasis	207/750 (27.6)	263/758 (34.7)
Irritability of airway mucosa	16/750 (2.1)	20/758 (2.6)
Serious adverse event (SAE) – no./total no. (%)		
Death	63/750 (8.4)	53/758 (7.0)
Was life-threatening	8/750 (1.1)	13/758 (1.7)
Involved or prolonged hospital length of stay	29/750 (3.9)	37/758 (4.9)
Involved persistence of significant disability or incapacity	0/750 (0.0)	0/758 (0.0)
Resulted in a congenital anomaly or birth defect	0/750 (0.0)	1/758 (0.1)

¶ See page 9 for the definitions of AE and SAE registration. There is a difference in the denominator of the patients for both groups compared to the outcome tables because patients were observed for AEs and SAEs up until a potential consent withdrawal. In case consent was withdrawn and no event was registered, the patient was registered as having no events and included in this table. No Serious Adverse Reactions (SAR) or Suspected Unexpected Serious Adverse Reactions (SUSAR) were registered.

eTable 10. Distribution of Oxygen Interventions Among the Post-Randomization Excluded Patients ¶		
	Restrictive oxygen group (N=750)	**Liberal oxygen group (N=758)**
Exclusion after randomization – no./total no. (%) (N=130)	55/750 (45)	67/758 (55)
Secondary exclusion – no./total no. (%) (N=341) *	174/750 (51)	165/758 (49)

¶ Detailed numbers on the post-randomization excluded patients from Figure 1 in the main manuscript.
* Two patients had missing data on the randomized oxygen intervention, which is the reason for the discrepancy between N=341 and the numbers in the table.

eTable 11. Best/Worst Case Scenario: Primary and Key Secondary Outcomes				
	Restrictive oxygen group	**Liberal oxygen group**	**Odds ratio (95% CI)**	***P* value**
Primary outcome				
Missing not counted as event: Death and/or major respiratory complications – no./total no. (%)	118/750 (15.7)	121/758 (16.0)	0.99 (0.74 to 1.33)	0.96
Missing counted as event: Death and/or major respiratory complications – no./total no. (%)	135//750 (18.0)	155/758 (20.4)	0.85 (0.65 to 1.11)	0.24
Key secondary outcomes				
Missing not counted as event: Death – no./total no. (%)	63/750 (8.4)	53/758 (7.0)	1.28 (0.86 to 1.92)	0.22
Missing counted as event: Death – no./total no. (%)	80/750 (10.7)	87/758 (11.4)	0.93 (0.68 to 1.30)	0.70
Missing not counted as event: Major respiratory complications – no./total no. (%)	65/750 (8.7)	78/758 (10.3)	0.82 (0.57 to 1.16)	0.26
Missing counted as event: Major respiratory complications – no./total no. (%)	82/750 (10.9)	112/758 (14.8)	0.70 (0.52 to 0.95)	0.02

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